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RESEARCH ARTICLE
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Global Eddy Subduction Carbon Pump From Argo Floats

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Key Points:

- The eddy subduction pump is evaluated globally from Argo float data, with 1,333 particulate organic carbon subduction events identified
- The eddy subduction pump peaks during springtime in hotspots of eddy kinetic energy, linked to mixed layer and symmetric instabilities
- The estimated global particulate organic carbon export flux below 200 m by the eddy subduction pump is 0.05 [<0.01–0.28] Pg C yr^{-1}

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract The eddy subduction pump, part of the biological pump, transports carbon-rich surface filaments downward via kilometer-scale turbulence in frontal regions, over days to weeks. Because of its spatial and temporal scales, this pump has been challenging to observe and quantify. Prior studies assessed the eddy subduction pump but were geographically limited. We introduce a detection algorithm that identifies subduction events using colocated anomalies in apparent oxygen utilization, absolute salinity, and optical backscattering. We apply it to 126,591 biogeochemical Argo profiles collected by 941 floats. The algorithm identifies 1,333 carbon subduction events concentrated in springtime hotspots of high eddy kinetic energy in the Southern Ocean and the subpolar North Atlantic. We estimate a global particulate organic carbon export flux below 200 m of 0.05 [<0.01–0.28] Pg C yr^{-1}. About two-thirds of this export occurs in the Southern Ocean (40%) and the North Atlantic (25%). Less than 25% of this flux enters water masses that remain isolated from the atmosphere for at least 50 years. Including dissolved organic carbon yields a total organic carbon export of 0.09 [0.01–0.51] Pg C yr^{-1}, less than 5% of the total organic carbon exported by the biological pump and an order of magnitude smaller than previous estimates. Although difficult to resolve in Earth system models, the eddy subduction pump appears to be a secondary term in global carbon budgets.

Plain Language Summary Tiny algae called phytoplankton use sunlight to turn dissolved carbon dioxide into organic matter. Some of this carbon leaves the surface ocean, is transported into the interior ocean, and eventually circles back to the surface ocean, decades to millennia later. This process is called the biological carbon pump and consists of multiple pathways. One pathway, the eddy subduction pump, occurs when swirling kilometer-scale eddies along ocean fronts push filaments of carbon-rich surface water downward. Because these events are small and short-lived, they have been difficult to observe. We use measurements from 941 robotic floats that drift, dive, and resurface, much like weather balloons for the ocean, to detect these events worldwide. Most events occur in the Southern Ocean and the subpolar North Atlantic, especially in spring when surface waters are rich in organic carbon. The eddy subduction pump transports less than 5% of the ocean's total organic carbon export, and therefore accounts for a small fraction of the total amount of carbon transported by the ocean.

1. Introduction

The biological carbon pump transfers carbon to the ocean interior, maintaining a vertical gradient in dissolved inorganic carbon (DIC), with higher concentrations at depth than at the surface. By reducing surface DIC, the biological pump maintains the ocean's capacity to absorb atmospheric carbon (Volk & Hoffert, 1985), regulating atmospheric carbon dioxide concentrations on seasonal to millennial timescales (DeVries, 2022). The surface to deep carbon gradient is sustained by all mechanisms that transport dissolved and particulate carbon of biological origin from the ocean surface to the interior (Frenger et al., 2024). The biological pump exports 10 ± 5 Pg C yr^{-1} of particulate organic carbon from the surface to the deep ocean (Boyd et al., 2019; Nowicki et al., 2022; Wang et al., 2023). Once viewed solely as the gravitational sinking of particles (Falkowski et al., 1998), the biological pump is now understood as a set of interdependent biophysical pumps operating across hours to decades and spanning kilometer to basin scales, including the physical transport of carbon through submesoscale eddies, large-scale circulation, the mixed layer ventilation and its restratification, and active transport through vertical migration of organisms (Boyd et al., 2019; Nowicki et al., 2022; Poupon et al., 2025; Siegel et al., 2023).

Among the various pumps that make up the biological carbon pump, the eddy subduction pump remains one of the least well understood, with estimates of its magnitude ranging from 0 to 2 Pg C yr^{-1} (Boyd et al., 2019). The eddy subduction pump is defined as a physically driven component of the biological carbon pump, involving the transport of carbon-rich surface waters by the large vertical velocities associated with physical instabilities at

submesoscale (0.1–10 km) fronts (Lévy et al., 2024; Omand et al., 2015). Observations with gliders and autonomous Argo floats have detected the eddy subduction pump in specific oceanic regions, for example, the North Atlantic, the Southern Ocean and the Kuroshio (M. L. Chen & Schofield, 2024; S. Chen et al., 2021; Lacour et al., 2019; Llort et al., 2018; Omand et al., 2015). Yet, a global evaluation remains elusive because the eddy subduction pump is inherently episodic and localized, making it difficult to capture with sparse observational coverage and challenging to extrapolate beyond regional studies. In addition, most previous studies have relied on region-specific detection algorithms, which were tailored to local oceanographic conditions and therefore not directly transferable to other regions (e.g., S. Chen et al., 2021; Lacour et al., 2023; Llort et al., 2018; Omand et al., 2015). As a result, attempts at a global quantification of the eddy subduction pump have been highly uncertain: early estimates extrapolated North Atlantic springtime observations to the global ocean and all seasons, suggesting an export of up to ~2 Pg C yr^{-1}, about 20% of the biological carbon pump's export (Boyd et al., 2019; Omand et al., 2015). However, more recent estimates based on observations in the Southern Ocean and an idealized North Atlantic model place its contribution below 19% and 5%, respectively (Llort et al., 2018; Resplandy et al., 2019). More recently, a global estimate of the physical injection pump of particulate organic carbon—where the eddy subduction contribution is inferred as the residual between the total injection pump and the large-scale components using a four-dimensional particulate organic carbon budget—yields an eddy subduction export of ~0.06 Pg C yr^{-1} (Bellacicco et al., 2025).

Uncertainty in the magnitude of the eddy subduction pump arises from its strong spatial and temporal variability, as well as from methodological differences in how subduction is detected, aggregated, and extrapolated. Seminal work by Omand et al. (2015) suggested that the eddy subduction pump could operate across broad regions in the mid to high latitudes, but subsequent regional studies indicate that the eddy subduction pump is instead concentrated in a limited number of hotspots shaped by distinct dynamical characteristics such as proximity to topographic features (Llort et al., 2018), high kinetic energy at frontal zones (S. Chen et al., 2021; Lacour et al., 2023) or seasonally deep mixed layers (A. R. Johnson & Omand, 2021). Temporally, the picture is equally complex: while some studies found enhanced activity of the eddy subduction pump in late winter to early spring (S. Chen et al., 2021; Erickson & Thompson, 2018), others found that the eddy subduction pump peaks in late spring or summer (A. R. Johnson & Omand, 2021; Llort et al., 2018), and still others detected no clear seasonality at all (Lacour et al., 2023). These conflicting findings reflect the limitations of regionally restricted studies and emphasize the need for a multi-year global assessment of the eddy subduction pump. However, such an assessment has so far been hindered by the absence of a globally applicable detection algorithm as existing approaches relied on region-specific criteria tailored to local dynamics (M. L. Chen & Schofield, 2024; A. R. Johnson & Omand, 2021; Lacour et al., 2023; Llort et al., 2018). Overcoming this technical gap is essential to enable a robust quantification of the eddy subduction pump magnitude, seasonality, and contribution to the biological carbon pump.

The mechanisms driving the eddy subduction pump remain debated. Eddy subduction events are by-products of the restratification of submesoscale fronts (Lévy et al., 2024; Taylor & Thompson, 2023), which in turn depend on both the processes that generate fronts and the instabilities that restratify them. Fronts may emerge either because of spontaneous frontogenesis or forced frontogenesis (Blumen, 2000; Hoskins & Bretherton, 1972; Shakespeare & Taylor, 2013), while front restratification is linked to instabilities such as symmetric instabilities (Bachman et al., 2017; Thomas et al., 2013), mixed layer instabilities (Boccaletti et al., 2007; Fox-Kemper et al., 2008), or a conjunction of both (Stamper & Taylor, 2017). Mixed layer instabilities are strongly modulated by the mixed layer depth (MLD), while symmetric instabilities depend on the upper ocean water column stability and vertical stratification, which can be evaluated using the squared Brunt-Väisälä frequency (N^2). A global assessment of the eddy subduction pump would therefore also provide an opportunity to examine how dynamical conditions, such as mixed layer depth and water column stratification, shape the occurrence of the eddy subduction pump, shedding light on the mechanisms at play.

In this study, we provide a global quantification of the eddy subduction pump and study its spatial and temporal variability. We build upon prior work that used subsurface anomalies in data collected by autonomous platforms, such as gliders and Argo floats (Llort et al., 2018; Omand et al., 2015). We extend this approach by applying an algorithm that detects signatures of the eddy subduction pump to the global biogeochemical Argo float database (Bittig et al., 2019). In the following, Section 2 presents the global biogeochemical Argo database, the global detection algorithm and the statistical methods used to analyze the data set. In Section 3, we analyze the 1,333

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detected carbon subduction events to identify spatial hotspots and assess the seasonality of the eddy subduction pump, as well as evaluate its potential physical and biological drivers. We estimate the probability of a carbon subduction event across the world and the mean particulate organic carbon content of carbon subduction events to derive a global estimate of carbon export by the eddy subduction pump. Finally, Section 4 discusses how our global assessment of the eddy subduction pump confirms or challenges previous regional studies, and we discuss the small estimated range for the eddy subduction pump (0.05 [<0.01–0.28] Pg C yr^{-1}) and its implications in the context of global carbon pump estimates.

2. Materials and Methods

2.1. Global Biogeochemical Argo Database

We used the 941 floats from the global biogeochemical Argo database (Bittig et al., 2019; Wong et al., 2020), equipped with sensors measuring salinity, temperature, oxygen, and particle backscattering at 700 nm (bbp_{700}). The data set spans from May 2010 to May 2024. We retrieved the float metadata and individual profiles using the OneArgo-R toolbox (Cornec et al., 2022), which provides access to the global index of available biogeochemical floats. The 941 selected floats yielded a total of 126,591 profiles, an order of magnitude more than the combined number of profiles analyzed in prior regional studies of the eddy subduction pump (M. L. Chen & Schofield, 2024; S. Chen et al., 2021; A. R. Johnson & Omand, 2021; Lacour et al., 2023; Llort et al., 2018). For each profile, we detected signatures of the eddy subduction pump using salinity, apparent oxygen utilization, and particle backscattering at 700 nm to detect subduction events (see Sections 2.2). Additionally, for subsequent analysis, we computed the maximum Brunt-Väisälä frequency across the water column as a proxy for vertical stratification. We also estimated the mixed layer depth using the standard density threshold method, defined as the depth where the potential density exceeds the surface value by more than 0.03 kg m^{-3} (de Boyer Montégut et al., 2004; Kelley et al., 2024).

The spatial coverage of the selected floats is uneven: only 33% of 1° × 1° ocean grid cells contain at least one profile, meaning that even at the mesoscale, approximately two-thirds of the global ocean remain unsampled, see Figure S1 in Supporting Information S1. Data coverage is densest in regions such as the North Atlantic, the Southern Ocean, the northern Indian Ocean, and the Kuroshio Extension, where many 1° × 1° grid cells contain more than 25 profiles, and in some cases, over 100. In contrast, large areas are undersampled, including the southwestern Gulf Stream, most of the Pacific (particularly the tropics), parts of the Kuroshio Extension, the South Atlantic, and the southern Indian Ocean, where grid cells generally contain fewer than 5 profiles. Notably, because of sea ice, the Arctic Ocean lacks Argo profiles entirely.

2.2. Subduction Events Definition

Building on previous methods for detecting subduction events (M. L. Chen & Schofield, 2024; S. Chen et al., 2021; K. S. Johnson et al., 2017; Lacour et al., 2023; Llort et al., 2018), we define two types of events that can be identified using the Argo float database: *physical subduction events* and *carbon subduction events*. *Physical subduction events* are defined as colocated subsurface negative anomalies in apparent oxygen utilization (AOU), defined as the difference between oxygen solubility and measured oxygen, and positive or negative anomalies relative to the vertical background in salinity (S), but not necessarily an anomaly in bbp_{700} (Figure 1). *Carbon subduction events* are a subset of physical subduction events, marked by the same AOU and salinity anomalies but with an additional positive anomaly in bbp_{700}, a proxy for particulate organic carbon (POC). This classification reflects the idea that a parcel of near-surface water injected into the ocean interior via subduction will carry surface-like properties, namely, elevated salinity, higher oxygen concentrations (i.e., lower AOU), and possibly elevated POC relative to surrounding waters. Such intrusions appear as local extrema relative to the background vertical structure (Figure 1).

While previous studies have used spiciness, which is a weighted combination of temperature and salinity (McDougall & Krzysik, 2015) to detect water mass intrusions (M. L. Chen & Schofield, 2024; S. Chen et al., 2021; Llort et al., 2018), we opt for salinity instead. Salinity is a conservative tracer and has been shown to perform as well as, or better than, spiciness in identifying water masses (McDougall et al., 2021; Tailleux, 2021). In the context of subduction event detection, salinity proves to be more robust because it is unaffected by temperature diffusion. Spiciness anomalies, by contrast, may be diminished if the associated temperature signal

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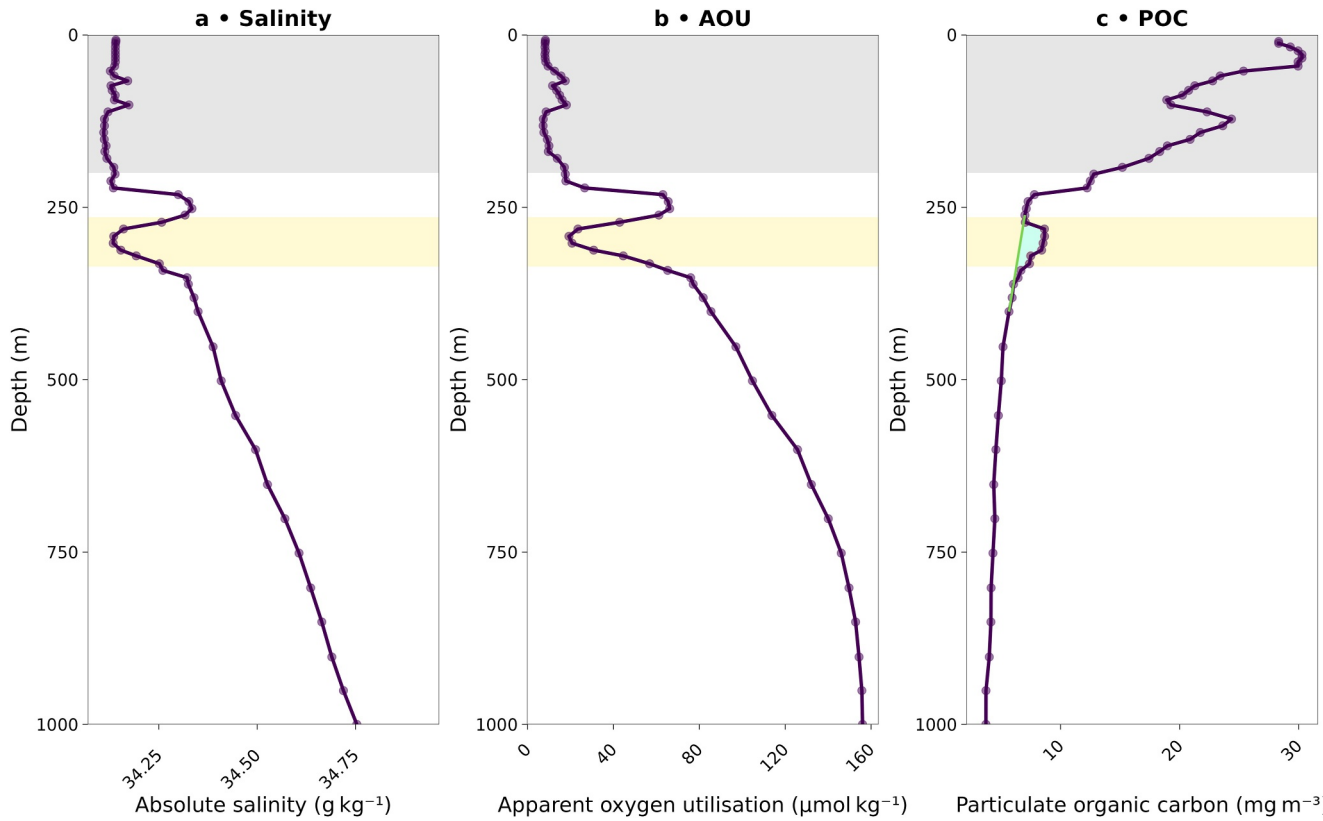

Figure 1. Vertical profiles from one Argo float in the Southern Ocean: (a) Salinity, (b) Apparent Oxygen Utilization (AOU) and (c) POC derived from backscatter propagation at 700 nm (Galí et al., 2022). Yellow shading shows the carbon subduction event detected by the algorithm using salinity, AOU and POC anomalies. Gray shading shows the top 200 m of the water column skipped by the detection algorithm (no detection in top 200 m). Green line and light blue shading shows the depth-integrated POC anomaly ($POC_{int} = 121\ mg/m^2$ in this case). Data are from float WMO5904676 cycle 58 on 5 October 2017 at 49.3°S, 110°E.

dissipates before the subduction event is captured by the float. A sensitivity analysis applying the same detection algorithm using spiciness instead of salinity resulted in substantially fewer detected events (approximately 7,000 using spiciness vs. approximately 9,900 using salinity), indicating that spiciness can miss subduction events due to the attenuation of temperature-compensated anomalies. An example of this effect is shown in Figure S2 in Supporting Information S1. Thus, using salinity reduces the likelihood of false negatives (i.e., cases where subduction events occur but go undetected). We note that in freshwater systems lacking vertical salinity gradients (e.g., lakes), alternative conservative tracers would be required.

2.3. Subduction Events Detection Algorithm

Prior work detecting subduction events was tailored to specific ocean regions (e.g., the Southern Ocean, the Kuroshio extension, and the North Atlantic) and could not be used at the global scale (M. L. Chen & Schofield, 2024; S. Chen et al., 2021; Llort et al., 2018). Here, we expand this work and develop a statistical algorithm to detect the footprints of subduction events that works for the global ocean (Keutgen De Greef, 2025). The algorithm scans each Argo profile vertically and, at each depth below 200 m, looks for negative anomalies in AOU, positive anomalies in bbp_{700}, and positive or negative anomalies in absolute salinity. Note that we do not use the top 200 m to detect physical and carbon subduction events because the primary goal of this study is to evaluate the impact of carbon subduction on the eddy pump below a fixed depth horizon of 200 m, which is a depth that is deeper than the seasonally varying mixed layer depth in most of the ocean (de Boyer Montégut et al., 2004).

The main difference with prior work is that we replace the cutoffs tailored to specific ocean regions with a robust residuals-based approach that can be applied globally. Anomalies in AOU, salinity, and bbp_{700} are identified by comparing each measurement to a robust estimate of the background vertical trend. In practice, we compute a 3-bin moving average and subtract a robust 9-bin trimmed mean to obtain the anomaly signal. The trimmed mean is

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an estimate of the mean of a random variable which is less sensitive to outliers than the arithmetic mean (Huber, 2004). The difference between the 3-bin moving average and the 9-bin trimmed mean is then standardized using a robust estimator of variance, the normalized interquartile range, an estimator of the standard deviation of a random variable which is also less sensitive to outliers than the sample standard deviation (Rousseeuw & Hubert, 2018). This procedure yields robust standardized residuals.

Observations with a standardized residual exceeding approximately 2 σ (97.5th percentile of a standard normal distribution) for AOU and salinity and 1 σ (68th percentile) for bbp_{700} are flagged as *carbon subduction events* (Figure 1). The lower threshold for bbp_{700} is used because of its relatively higher noise level. Data points meeting the residual criteria for AOU and salinity but not for bbp_{700} are classified as *physical subduction events*. Lastly, we verify that the finite-difference approximation of the first derivative changes sign within ±100 m of the detected level; this ensures that the detected level represents local extremes in AOU, salinity, and bbp_{700}, thereby improving the algorithm's specificity (Keutgen De Greef, 2025).

We tuned the detection algorithm toward high sensitivity to minimize missed events, prioritizing the reduction of false negatives over the reduction of false positives. In practice, it is preferable to flag many potential events, including false positives that can later be discarded by inspection, rather than risk missing true subduction events. Of the 9,979 anomalies flagged by the algorithm, manual inspection confirmed 4,377 true positives and rejected 5,602 false positives according to the following criteria: colocated AOU and salinity peaks clearly identifiable by eye, situated below the mixed layer, with a vertical extent smaller than 200 m. To quantify the false-negative rate, we randomly selected 1,000 profiles from nine floats and manually classified every profile they recorded. The algorithm correctly detected 92% of genuine physical subduction anomalies, corresponding to a false-negative rate of about 8%, which we considered acceptable. Thresholds were chosen after exploratory tests showed that stricter cutoffs (e.g., $>2\sigma$) eliminated many genuine anomalies, whereas looser ones ($<1\sigma$) produced an unmanageable number of false positives. In the end, the algorithm, combined with manual verification, yielded 4,377 true physical subduction events and 1,333 true carbon subduction events.

2.4. Probability of Subduction and Global Statistical Interpolation

The detection algorithm and the subsequent manual verification give the spatial and temporal location of physical and carbon subduction events and reflect the uneven coverage of the Argo database. Thus, more subduction events are detected in well-sampled regions (Figures 2a and 2b). To take this sampling bias into account, we perform a statistical interpolation that estimates a smooth, large-scale $\mathcal{O}$(1,000 km) probability field of physical and carbon subduction, allowing us to more reliably identify regions of enhanced subduction activity. We grid the data in $1° \times 1°$ grid cells, and consider the proportion p_i of y_i physical or carbon subduction events detected in grid cell i to N_i, the total number of Argo profiles in the grid cell i: $p_i = y_i/N_i$.

The random quantity p_i lends itself to a straightforward interpretation: in the limit where the number of Argo profiles in a grid cell is sufficiently large, this is the probability that an Argo float present in that grid cell captures a physical or carbon subduction event. However, the computed values p_i are still exposed to a bias, as grid cells with very few total numbers of Argo profiles (N_i) are likely to have an unrealistically high or low estimated p_i due to sampling variability. Therefore, we fit statistical generalized additive models (GAMs), with spherical splines to account for Earth's spherical geometry (Wahba, 1981), to estimate a smooth, large-scale probability field based on the observed proportions p_i. We refer to this interpolated average probability as $\hat{p}_i$. We account for bias arising from sparsely sampled grid cells by treating 1° grid cells containing fewer than five Argo profiles as missing data. In addition, we weight observation i by $\log(N_i)$, such that the fitted models give more weight to more reliable observations, that is with a higher number of Argo profiles in the grid cells. The GAMs are optimized to explain at least 60% of the model deviance (Hastie & Tibshirani, 1986), balancing model complexity and goodness of fit. We then interpolate the fitted model onto a finer 0.25° grid to produce smooth maps of the probability of physical subduction and of carbon subduction (Hastie et al., 2001; Wahba, 1981). Although the GAM model is fit onto a 0.25° evaluation grid, it does not imply that it resolves 0.25° features. The effective spatial resolution of the GAM model is set by the density of the Argo observations (Figure S1 in Supporting Information S1) and the spatial smoothness of the model (see Text S1 in Supporting Information S1 for details), not by the evaluation grid. The 0.25° grid is used for visualization and for comparison with model-derived fields. A comparison between proportion maps at 5° and 1° resolution and the GAM-derived probability field evaluated on the 0.25° grid is shown

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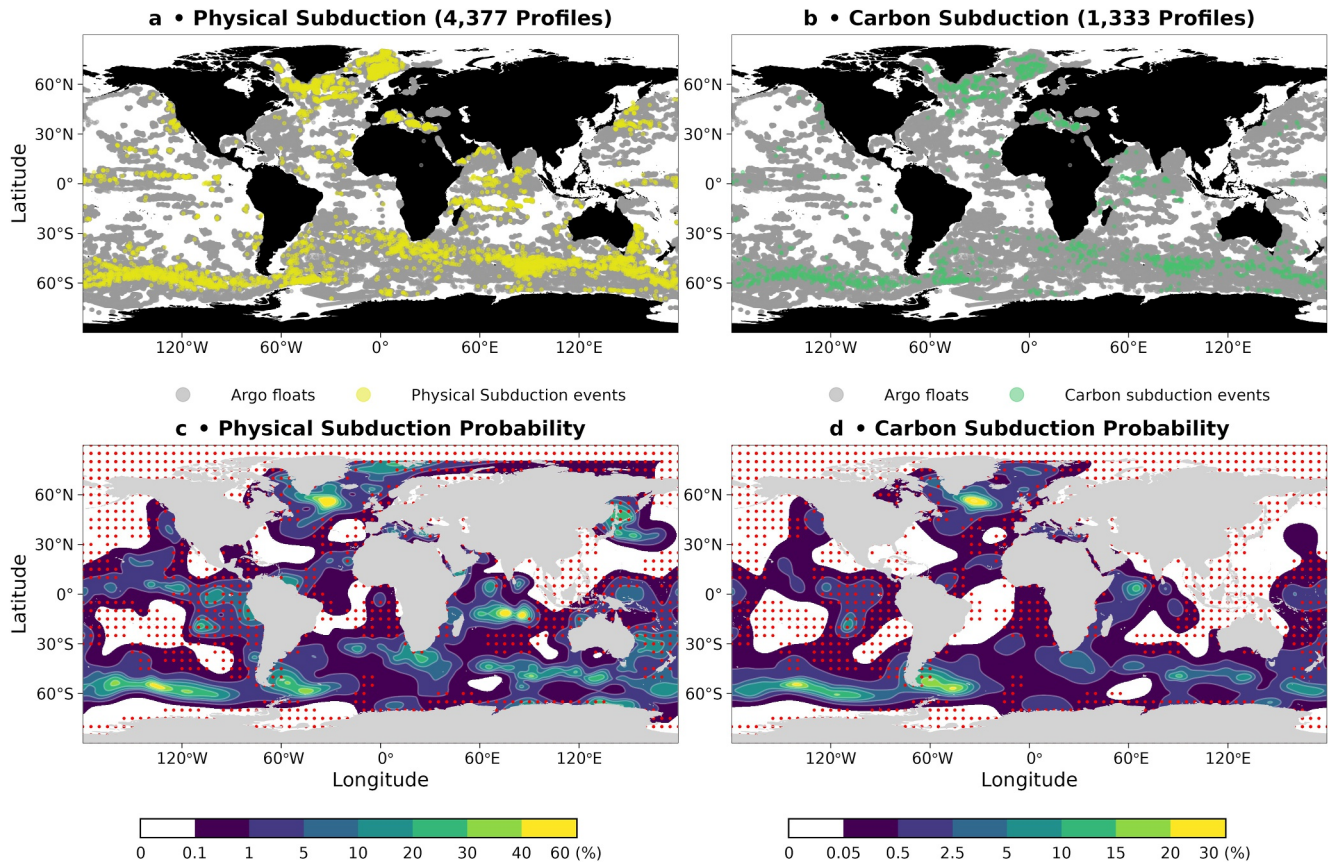

Figure 2. Spatial distribution of detected physical subduction events and carbon subduction events, raw observations and interpolated. (a) Argo profiles where physical subduction events were detected (yellow dots) and where no physical subduction events were detected (gray dots). (b) Argo profiles where carbon subduction events were detected (green dots) and where no physical subduction events were detected (gray dots). (c, d) Maps of the probability that an Argo float detects a physical subduction event (c) and a carbon subduction event (d). Probabilities are estimated from generalized additive models (GAMs). Red stippling indicates 5° grid cells with no Argo profiles, indicating insufficient spatial coverage.

in Figure S3 in Supporting Information S1. More details on the estimation of GAMs are in Text S1 in Supporting Information S1.

2.5. Eddy Kinetic Energy and Turnover Time Data Sets

To assess the relationship between subduction events and mesoscale eddy activity, we used both balanced eddy kinetic energy (EKE), reflecting low-frequency, geostrophically balanced mesoscale eddies and fronts and unbalanced eddy kinetic energy, reflecting high-frequency, ageostrophic internal tides and internal-gravity/near-inertial waves (Qiu et al., 2018). Balanced and unbalanced EKE measurements are derived from high-resolution 1/48° MITgcm simulations by Qiu et al. (2018). This resolution is sufficient to resolve the first baroclinic deformation radius globally, including at high latitudes and in relatively low-energy regions where the deformation radius is smallest (Hallberg, 2013). As a result, the EKE diagnostics capture mesoscale eddies and frontal strain across all ocean regimes, and not only in energetic western boundary currents or the Southern Ocean.

Balanced and unbalanced eddy kinetic energy are defined for a given horizontal wavenumber k_h as

$$\mathrm{KE}_{\mathrm{bal}}(k_h) = \int_{\omega_{yr}}^{\omega_{cr}} \hat{E}(k_h,\omega)\,d\omega, \quad \mathrm{KE}_{\mathrm{unbl}}(k_h) = \int_{\omega_{cr}}^{\omega_{Ny}} \hat{E}(k_h,\omega)\,d\omega, \tag{1}$$

where $\mathrm{KE}_{\mathrm{bal}}(k_h)$ represents the balanced kinetic energy integrated in wave number space from the yearly frequency (ω_{yr}) to the critical frequency (ω_{cr}) and $\mathrm{KE}_{\mathrm{unbl}}(k_h)$ corresponds to the unbalanced kinetic energy integrated from the critical frequency (ω_{cr}) to the Nyquist frequency (ω_{Ny}). The critical frequency is defined as the

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minimum of either the local tenth vertical-mode internal gravity wave dispersion frequency or the permissible tidal frequencies (Qiu et al., 2018).

The high-resolution EKE fields were spatially averaged onto a 0.25° grid to facilitate comparison with the GAM-interpolated map of the probability of physical subduction.

To quantify the association between EKE and subduction probability, we fitted generalized linear models (GLMs) across two latitudinal bands: the tropics (30°S to 30°N) and the extratropics (latitudes greater than 30°N or 30°S). Further methodological details are provided in Text S2 in Supporting Information S1.

We also used the dissolved inorganic carbon reemergence time fields of Siegel et al. (2021). The reemergence time is defined as the time necessary for the water masses to return to the surface. The data product is provided on a 2° horizontal grid with 48 vertically irregular layers. We applied nearest-neighbor interpolation to collocate these fields with our grid and, for each season, we estimated the fraction of POC exported by the eddy subduction pump below 200 m that is expected to remain out of contact with the atmosphere for at least 50 years. Following Frenger et al. (2024) and Visser (2025), we avoid the term "sequestration timescale" for DIC and instead use "reemergence time" to emphasize that this metric reflects ventilation/return to the surface rather than net offset of anthropogenic emissions.

2.6. Quantification of Carbon Export From Carbon Subduction Events

We estimate the contribution of each carbon subduction event to the eddy subduction pump by computing the local flux below 200 m of particulate organic carbon (POC) attributable to these events for each season and each 0.25° grid cell. This estimation proceeds in several steps. First, for each detected carbon subduction event, we convert bbp_{700} profiles to POC concentrations (in mg C m^{-3}) using the biome- and depth-dependent empirical formulation of Galí et al. (2022), which yields conservative POC estimates within the 200–1,000 m depth range (see Text S4 in Supporting Information S1 for details).

We then estimate the POC anomaly associated with the event by computing the depth-integrated deviation from a background profile (Figure 1c). Specifically, we define the anomaly as the difference between the observed POC concentration, $POC^{obs}(z)$, and a linear background profile, $POC^{lin}(z)$, interpolated between the nearest local extrema above (z_1) and below (z_2) the anomaly within a 300 m window

$$POC_{int} = \int_{z_1}^{z_2} \left(POC^{obs}(z) - POC^{lin}(z)\right) dz \quad \left[\text{mg C m}^{-2}\right]. \tag{2}$$

Dividing the inventory of excess particulate carbon contained in a subducted filament POC_{int} by the export horizon H converts that inventory to an average concentration anomaly over the depth range where export of the eddy subduction pump is presumed to occur.

To obtain spatially continuous maps, we fit a generalized additive model to POC_{int}, yielding seasonal interpolated fields of the depth-integrated POC anomaly across the 0.25° grid (see Figure S4 in Supporting Information S1). We consider POC_{int} to be the inventory of excess particulate carbon contained in a subducted filament. This approach parallels the generalized additive model interpolation of the probability of carbon subduction events, p presented in Section 2.4 (see Text S1 in Supporting Information S1 for details).

Next, we use the interpolated fields of integrated POC anomaly (POC_{int}) to estimate the local daily POC flux, Q, attributable to the eddy subduction pump:

$$Q = POC_{int} \times \frac{W}{H} \times \hat{p} \quad \left[\text{mg C m}^{-2}\text{d}^{-1}\right], \tag{3}$$

where W is the characteristic vertical velocity of subduction (in m d^{-1}), $H = 200$ m is the assumed depth of the export horizon, and $\hat{p}$ is the interpolated seasonal probability of detecting a carbon subduction event. The average concentration anomaly POC_{int} is multiplied by the characteristic vertical velocity of subduction W and this product yields the advective flux associated with a single event. Lastly, $\hat{p}$ scales this single-event flux by the seasonal probability of a carbon subduction event, producing a local daily flux for each grid cell and each season.

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Additionally, we account for the possibility that the integrated POC anomaly (POC_{int}) observed by Argo floats may underestimate the total amount originally subducted, due to remineralization during the time lag between subduction and detection. Following A. R. Johnson and Omand (2021), who reported subduction anomalies persisting for up to 3 months, we therefore apply a high-bound exponential correction assuming an e-folding decay timescale of 90 days. Specifically, we multiply POC_{int} by e to obtain an upper-bound estimate of the original anomaly such that Q may be upper-bounded as

$$Q^{+} = e \times \mathrm{POC}_{\mathrm{int}} \times \frac{W}{H} \times \hat{p} \quad \left[\mathrm{mg\ C\ m^{-2} d^{-1}}\right]. \tag{4}$$

The impact of this correction is also included in the sensitivity analysis shown in Figure S6 in Supporting Information S1.

In addition, using the interpolated reemergence time product of Siegel et al. (2021), we estimate for each seasonal flux Q and its upper-bound Q^{+}, the mean fraction of the exported carbon that will be trapped in water masses that remain out of contact with the atmosphere for at least 50 years, this yields $Q_{50\ \mathrm{yrs}}$ and $Q^{+}_{50\ \mathrm{yrs}}$.

Lastly, the global annual export attributable to the eddy subduction pump and expressed in Pg C yr^{-1} is then obtained by integrating Q or its upper-bound Q^{+} over the ocean's surface and over seasons. In addition, the fraction of that global annual export that happens in water masses that remain sequestered for more than 50 years is also obtained by integrating $Q_{50\ \mathrm{yrs}}$ and its upper-bound $Q^{+}_{50\ \mathrm{yrs}}$.

The limitations and assumptions behind this method are as follows.

- Constant depth of the export horizon (H). The assumption of constant depth of the export horizon H assumes that the relevant export horizon of the eddy subduction pump is 200 m. This choice is motivated by the fact that 200 m lies below the climatological maximum mixed layer depth over most of the global ocean (de Boyer Montégut et al., 2004), ensuring that detected carbon subduction events are clearly decoupled from the surface mixed layer and unlikely to be rapidly re-entrained. However, shallower export in the tropics (lower H) or deeper export in subpolar gyres (higher H) would lead to under- or over-estimation of the eddy subduction flux, respectively.
- Uniform characteristic velocity (W). Here, we use $W = 200$ m d^{-1} as a representative mid-range value, consistent with observational estimates of submesoscale dynamics (Torres et al., 2024; Zhu et al., 2024), but also include the sensitivity to a plausible range for vertical velocities between 20 and 500 m d^{-1} (Stamper & Taylor, 2017) to provide a first-order uncertainty envelope for our eddy pump flux estimate. The sensitivity of Q to W is shown in Figure S6 in Supporting Information S1.
- Event independence. We assume event independence in computing $\hat{p}$. However, in regions where subduction clusters in space or time, this assumption may inflate the effective frequency of individual carbon subduction events and therefore over-estimate of the local daily flux.
- Stationarity of the probability of carbon subduction within each season. In Equation 3, we multiply by $\hat{p}$, which is fixed for each season. However, the probability of carbon subduction is likely to be highly dependent on intraseasonal dynamics such as wind speed or sea surface temperature and salinity, affecting front formation and eventually, subduction.
- Neglect of lateral advection and frontal interleaving. Isopycnal transport and water-mass interleaving in regions of strong fronts may occasionally produce subsurface spurious tracer anomalies in AOU and salinity that resemble vertical subduction when sampled by floats, potentially leading to false positives.
- Neglect of lateral diffusion. Lateral diffusion erodes tracer anomalies between subduction and detection, leading to potential false negatives.
- Potential exponential-decay of the POC signal with time (see Equation 4).

In the discussion section (Section 4), we discuss the impact of the limitations here-above on our assessment of the eddy subduction pump.

2.7. Statistical Metrics

We conducted hypothesis tests to assess the statistical significance of seasonal variability in both the frequency and the depth of subduction events. This step is essential given the uneven spatial and temporal sampling of the Argo float database.

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To test whether carbon and physical subduction events exhibit significant seasonality, we evaluated the null hypothesis that normalized monthly detection probabilities are uniformly distributed. For each month i, the normalized detection probability $\hat{p}_i$ is defined as

$$\hat{p}_i = \frac{p_i}{\sum_{k=1}^{12} p_k}, \tag{5}$$

where p_i is the raw monthly probability. Under the null hypothesis, $\hat{p}_i$ follows a uniform distribution $U(1/12)$, and deviations from uniformity were assessed using a χ^2 goodness-of-fit test. Because tests were applied separately to multiple ocean regions, we controlled the family-wise error rate using a Bonferroni correction, setting the adjusted significance threshold to $\alpha = 0.05/5 = 0.01$ (VanderWeele & Mathur, 2019).

To examine whether the vertical extent of subduction events also varies seasonally, we applied a Kruskal-Wallis test to the monthly distributions of subduction depths. The test was performed on both the full set of detected events and on the subset comprising the deepest 20% of events (i.e., the top quintile).

3. Results

3.1. Spatial Distribution of Physical and Carbon Subduction

Primary hotspots of physical subduction, where the annual probability of subduction exceeds 30%, are detected in the subpolar North Atlantic, the equatorial Indian Ocean and in the Southern Ocean, specifically, between the Subantarctic Front and the Southern ACC Front, downstream of the Drake Passage and of the Kerguelen Plateau, south of New Zealand, near the East Pacific Antarctic Ridge and in the Agulhas Current (Figures 2a and 2b). Secondary hotspots, where the annual probability of physical subduction is 10%–30%, are found in the Mediterranean Sea, the Kuroshio and California Currents. Secondary hotspots are also detected in the equatorial Pacific and equatorial Atlantic, but these areas are poorly sampled by Argo floats, limiting confidence in the magnitude and spatial extent of these features (red dots in Figures 2c and 2d).

Hotspots of carbon subduction are less numerous than hotspots of physical subduction, and are primarily concentrated at mid- to high-latitude in the Southern Ocean and subpolar North Atlantic (Figures 2f and 2g). Primary hotspots of carbon subduction, where the annual probability of carbon subduction is over 10% are detected in the North Atlantic and the Southern Ocean downstream of the Drake Passage, and along the Pacific-Antarctic Ridge, while secondary carbon subduction hotspots, with probabilities between 2.5% and 10%, are detected in the Agulhas Current, downstream of the Kerguelen Plateau, south of New Zealand, in the Indian Ocean, and in the Eastern Pacific. Like for physical subduction, the equatorial Eastern Pacific, Indian Ocean, and equatorial Atlantic also contain weak hotspots, but the confidence in these hotspots is lower due to limited sampling Argo coverage (red dots in Figures 2c and 2d).

3.2. Seasonality of Physical and Carbon Subduction

We evaluate the seasonality of physical and carbon subduction events at the regional scale, and find higher subduction probabilities and stronger seasonality in the Southern Ocean and the North Atlantic than in the North Pacific, the Northern Tropics and the Southern Tropics (barplots in Figure 3). In both the Southern Ocean and the North Atlantic, the probability of physical subduction rises from 4% to 5% in winter to about 7% in spring/summer, and the probability of carbon subduction doubles from 1% to 1.5% in winter to 3% in spring (peak in September-October in the Southern Ocean and April-May in the North Atlantic). The winter-to-spring increase in carbon subduction departs significantly from a uniform distribution in both regions (Southern Ocean: $\chi^2 = 28.5$, $p = 2.7 \times 10^{-3}$, North Atlantic: $\chi^2 = 35.9$, $p = 1.8 \times 10^{-4}$), while the seasonal increase in physical subduction is only statistically significant in the Southern Ocean (Southern Ocean: $\chi^2 = 40.3$, $p = 3.2 \times 10^{-5}$, North Atlantic: $\chi^2 = 19.4$, $p = 0.054$). In both regions, the spring peak in carbon subduction coincides with the seasonal mixed layer restratification and the seasonal rise in chlorophyll (blue and green lines in Figures 3f and 3g).

In contrast, in the North Pacific, Northern Tropics, and Southern Tropics, subduction probabilities are lower and seasonality is weaker. Across these regions, physical subduction probabilities remain between 1.5% and 3%, while carbon subduction is generally below 1% (average ~0.3%), with modest peaks in summer in the Northern

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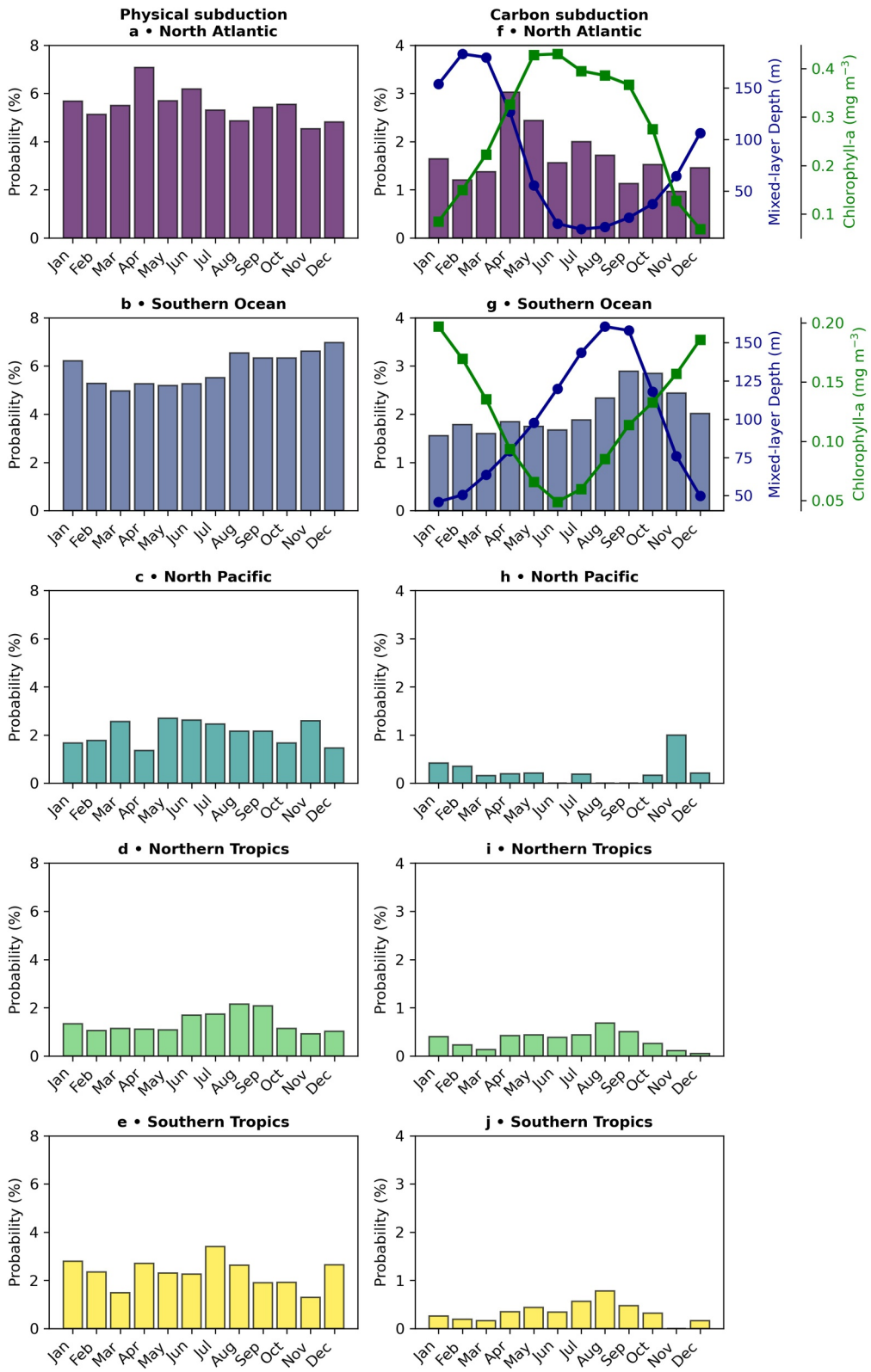

Figure 3.

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Tropics (August) and fall in the North Pacific (November). None of these seasonal fluctuations are statistically significant (North Pacific: $\chi^2 = 8.0$, $p = 0.71$; Northern Tropics: $\chi^2 = 18.9$, $p = 0.068$; Southern Tropics: $\chi^2 = 18.2$, $p = 0.077$). The lack of statistical significance should not be interpreted as proof that seasonality is absent, but that observations are too limited to draw firm conclusions about the presence or absence of seasonality.

3.3. Depth of Physical and Carbon Subduction Events

Globally, 70%–80% of physical and carbon subduction events lie between 200 and 500 m (Figure 4). Note that by design, no events are detected above 200 m. For both carbon and physical subduction, depth distributions vary across basins. Subduction events reach greater depths in the Southern Ocean, North Atlantic, and Southern Tropics, where about 30% of physical subduction events and 20% of carbon subduction events extend below 500 m (Figure 4). In contrast, events are generally shallower in the Northern Tropics and the North Pacific, with only about 20% and 7% of physical subduction events and 7% to none of carbon subduction events occurring deeper than 500 m (Figure 4). These observed patterns do not imply that carbon subduction cannot occur below 500 m in the Pacific, or that a third of physical subduction truly happens at depth below 500 m. Rather, sparse Argo coverage and the intrinsic rarity of deep carbon and physical subduction make such events easy to miss and the small number of detected events increases sampling error. Within each region we detect no statistically significant seasonal shift in either the mean event depth or the thickness of the distribution tail.

3.4. Potential Drivers of Physical Subduction

We first assess whether subduction aligns with the two pathways of frontogenesis: forced frontogenesis or spontaneous frontogenesis (see Section 1). To this end, we use the balanced and unbalanced EKE product as provided by Qiu et al. (2018) (see Section 2.5). Balanced EKE is a proxy for persistent mesoscale strain that drives forced frontogenesis, while unbalanced EKE (internal tides and near-inertial waves) energizes high-frequency shear that can either sharpen or weaken fronts episodically (Blumen, 2000; Hoskins & Bretherton, 1972; Shakespeare & Taylor, 2013).

In the extratropics (poleward of 30°), high balanced EKE is strongly associated with high probability of physical subduction (Figure 5a). Indeed, the odds of observing a subduction event rise sharply with balanced EKE, with a one-unit increase in the natural logarithm of balanced EKE, equivalent to multiplying balanced EKE by $e \approx 2.7$, boosts the odds of subduction by roughly 55% (odds ratio, OR $= 1.55$, p - value $< 10^{-15}$). In the tropics (equatorward of 30°), however, the odds of observing a physical subduction event declines with balanced EKE (OR $= 0.92$, p - value $< 10^{-15}$, Figure 5).

By contrast, high unbalanced EKE is strongly negatively associated with the probability of physical subduction in both the tropics and the extratropics. A one-unit increase in the natural logarithm of unbalanced EKE decreases the odds of subduction by roughly 21% (OR $= 0.79$, p - value $< 10^{-15}$) in the extratropics and 45% in the tropics (OR $= 0.55$, p - value $< 10^{-15}$), see Figure 5. The strong link between subduction and balanced EKE in the extratropics suggest that subduction occurs in regions of forced frontogenesis. In contrast, the reduction in subduction probability in regions of high unbalanced EKE suggest that internal tides and near inertial-waves suppress the occurrence of subduction (Blumen, 2000; Hoskins & Bretherton, 1972; Shakespeare & Taylor, 2013).

We also assess whether detected subduction events are more likely to be associated with mixed layer instabilities or symmetric instabilities. Deeper mixed layers are typically associated with mixed layer instabilities, while lower values of $N^2_{\max}$ are linked to symmetric instabilities. To evaluate how water-column structure modulates subduction, we compare the median MLD and the median $N^2_{\max}$ between profiles with and without detected subduction (Figures 5c and 5d). Vertical stratification (N^2) is significantly and robustly lower in profiles with subduction (median $= 2.2 \times 10^{-4}$, IQR$= 6.0 \times 10^{-4}$) than in profiles without subduction (median $= 4.8 \times 10^{-4}$, IQR $= 1.0 \times 10^{-3}$; Figures 5c and 5d). In contrast, the mixed layer depth distributions of the

Figure 3. Monthly probability of carbon subduction and physical subduction event in five ocean regions. (a, f) North Atlantic (latitude ≥30°N). (b, g) Southern Ocean (latitude ≥30°S). (c, h) North Pacific (latitude >30°N). (d, i) Northern Tropics (0° < latitude < 30°N). (e, j) Southern Tropics (0° ≤ latitude < 30°S). Regional monthly climatologies of chlorophyll from observation-based products (Melin, 2013) and mixed layer depth from biogeochemical Argo floats are overlaid on panels (f, g). In the North Atlantic and Southern Ocean, carbon subduction exhibits significant seasonality, peaking after mixed layer depth reaches its maximum but before chlorophyll's monthly climatology peaks (a, c).

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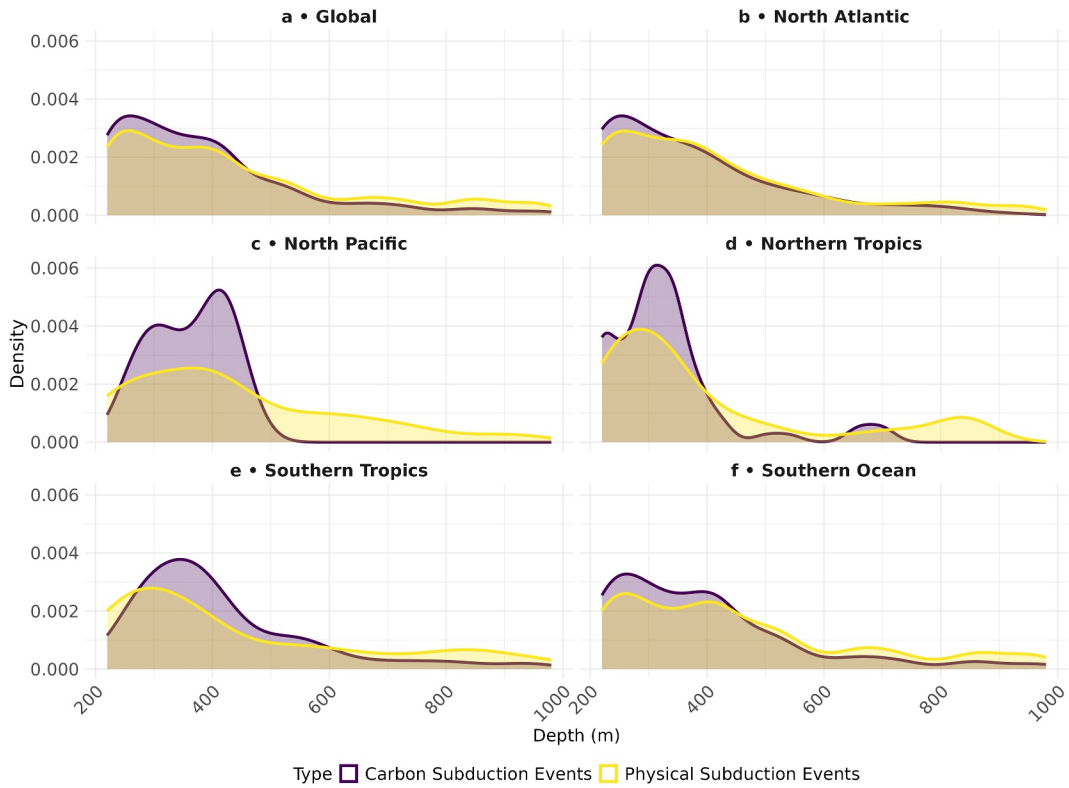

Figure 4. Depth of physical subduction events (yellow) and carbon subduction events (purple) for: (a) global ocean, (b) North Atlantic, (c) North Pacific, (d) Northern Tropics, (e) Southern Tropics, (f) Southern Ocean.

profiles with and without subduction are more similar. Profiles with subduction have somewhat deeper mixed layers (median 59.6 m, IQR 87.6 m) than those without subduction (median 44.3 m, IQR 67.1 m), but the overlap between the two distributions is large (see Figure 5).

Quantitatively, the Kolmogorov-Smirnov (KS) distance between the $N^2_{\max}$ distributions with and without subduction is twice that of MLD ($D = 0.22$ for $N^2_{\max}$ vs. $D = 0.11$ for MLD). We assessed whether this difference could arise from sampling error due to the limited number of profiles by bootstrapping the calculation of KS distances. In 100% of 1,000 bootstrapped samples, the KS distance for $N^2_{\max}$ exceeds that of MLD, allowing us to reject the null hypothesis that the observed difference is due to chance. This stronger separation between profiles with and without detected subduction in $N^2_{\max}$ compared to MLD indicates that weak stratification is more consistently associated with detected subduction events than deep mixed layers. In other words, statistical evidence suggests that subduction events might be more systematically associated with conditions favorable to symmetric instability than with those characteristic of mixed-layer instability. We note that this association alone does not identify the underlying physical instability behind the subduction as both symmetric and mixed-layer instabilities can occur simultaneously under weakly stratified conditions.

3.5. Global and Regional Export of the Eddy Subduction Pump

We evaluate the POC export below 200 m (see details in Section 2.6). Export fluxes are higher in the North Atlantic and the Southern Ocean where they reach around 5 [0.5, 50] mg C m^{-2} d^{-1}, while they generally remain below 1.0 [0.1,5] mg C m^{-2} d^{-1} in other regions (Figure 6b). These regional differences in fluxes arise from both the mean amount of POC subducted during each event (i.e., integrated POC anomalies) and the frequency of these events (i.e., the probability of carbon subduction), which are both higher in the North Atlantic and the Southern Ocean than elsewhere (Figure 2d and Figure S4 in Supporting Information S1). Yet we note systematic differences between the two basins. The North Atlantic is characterized by higher POC content per event (more than 150 mg C m^{-2} in boreal summer) than the Southern Ocean (100 mg C m^{-2} in austral summer) leading to stronger

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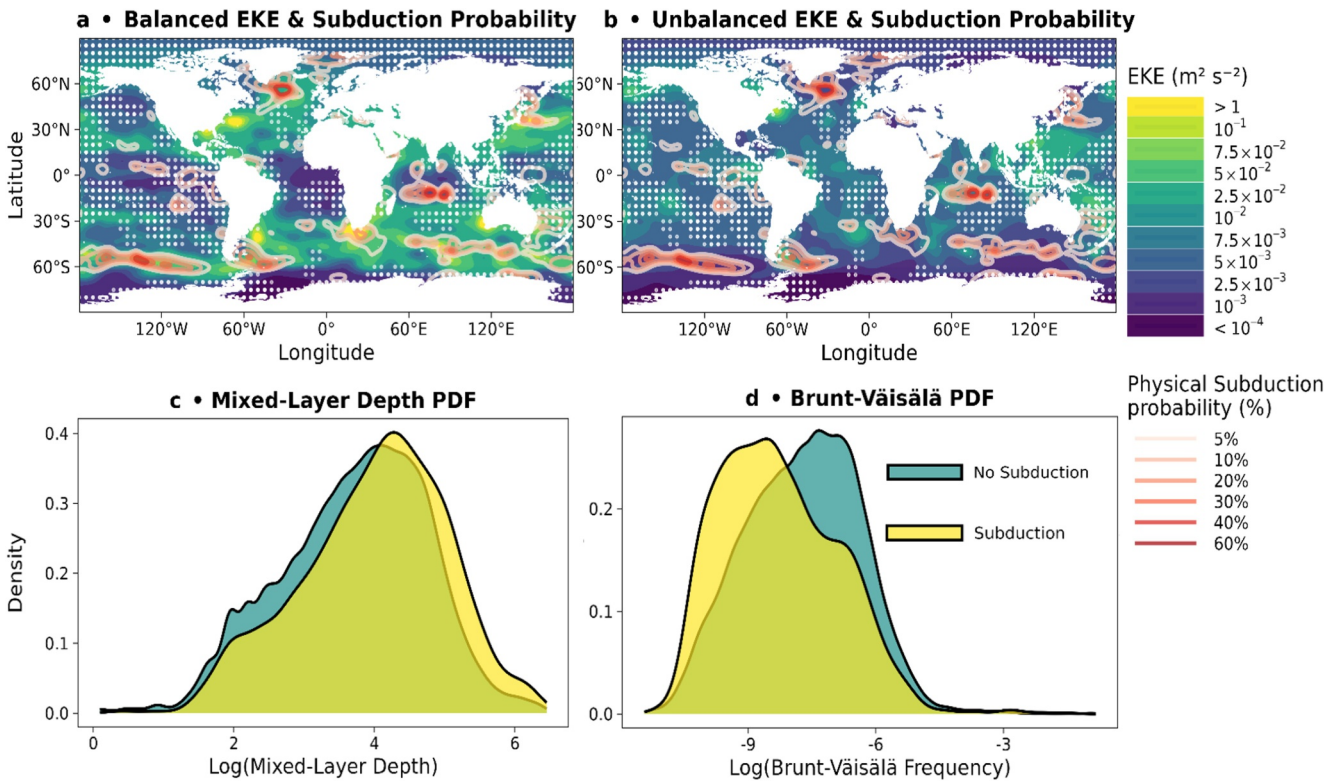

Figure 5. Physical drivers of subduction. (a, b) Balanced and unbalanced eddy kinetic energy (EKE) from (Qiu et al., 2018), overlaid with contour lines (red) of the probability that an Argo float captures a physical subduction event. Stippling indicates 5-degree grid cells with no Argo profiles, indicating insufficient spatial coverage. (c, d) Probability density functions of mixed layer depth (c) and Brunt-Väisälä squared frequency (d) for profiles with physical subduction (yellow curve) and without physical subduction (teal curve). The lower overlap in Brunt-Väisälä squared frequency probability density functions suggests that Brunt-Väisälä squared frequency differences better predict subduction than mixed layer depth.

local fluxes (Figure 6b and Figure S4 in Supporting Information S1). The Southern Ocean is characterized by more hotspots than the North Atlantic, leading to a greater export flux when aggregated at basin scale (Figure 6a).

Integrated globally, we find a POC export flux associated with the eddy subduction pump of 0.05 [<0.01–0.28] Pg C yr^{-1}. The central estimate assumes a velocity of 200 m d^{-1}, whereas the lower and upper estimates assume velocities of 20 m d^{-1} and 500 m d^{-1}, respectively (see Sections 2.6 and 4.5). The Southern Ocean contributes about 40% to this global flux (~0.02 [0.002–0.11] Pg C yr^{-1}), the North Atlantic about 25% (~0.01 [0.001–0.08] Pg C yr^{-1}), and the North Pacific together with the Tropics the remaining 35% (~0.018 [0.002–0.09] Pg C yr^{-1}; Figure 6b).

We evaluate the fraction of global export occurring in water masses estimated to remain out of contact with the atmosphere for more than 50 years, using the reemergence time product of Siegel et al. (2021). This fraction is estimated at 0.014 [0.001–0.07] Pg C yr^{-1} (Figure 6a). When restricting to waters isolated from the atmosphere for over 50 years, the relative contribution of the North Atlantic decreases from about 25% to 20% (~0.002 Pg C yr^{-1}), while the Southern Ocean remains near 40% (~0.006 Pg C yr^{-1}) and the North Pacific and Tropics rise from 35% to 40% (~0.006 Pg C yr^{-1}). The Mediterranean contributes around 3% of the total flux (~0.002 Pg C yr^{-1}), but less than 1% of the export that remains isolated for more than 50 years, pointing to shallower subduction in the Mediterranean compared to other regions (see Table S1 in Supporting Information S1).

4. Discussion

4.1. Carbon Subduction Concentrated in Hotspots

In this study, we used the global biogeochemical Argo database to provide a worldwide assessment of the eddy subduction pump. We developed a statistical framework to detect physical and carbon subduction events,

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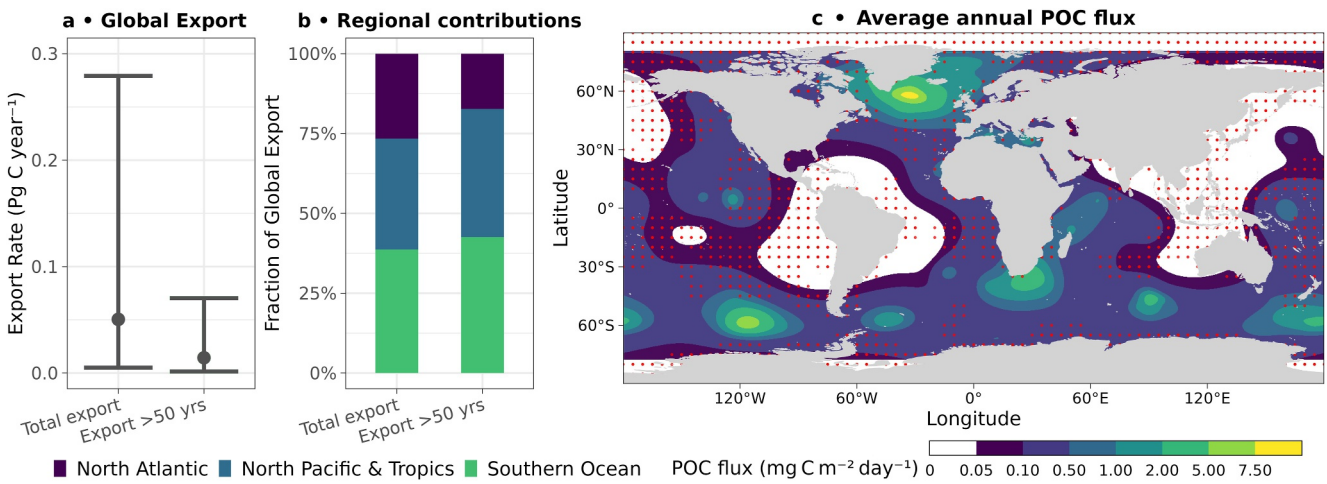

Figure 6. Global eddy subduction pump. (a) Global annual eddy subduction POC flux exported below 200 m: total flux (in blue) and flux exported in water masses that stay out of contact with the atmosphere for more than 50 years (in green). The central value corresponds to a subduction characteristic velocity of $W = 200\ \mathrm{m\ d^{-1}}$, while the lower and upper end values correspond to $W = 20\ \mathrm{m\ d^{-1}}$ and $W = 500\ \mathrm{m\ d^{-1}}$. (b) Regional relative contributions to annual eddy subduction POC flux: total and exported for more than 50 years. (c) Spatial distribution of total annual mean POC flux below 200 m assuming an average vertical velocity $W = 200\ \mathrm{m\ d^{-1}}$. Red stippling marks 5° grid cells without Argo profiles (insufficient coverage). Turnover timescales are taken from dissolved inorganic carbon-based estimates in (Siegel et al., 2021).

allowing us to estimate a global POC export flux of 0.05 [<0.01–0.28] Pg C $\mathrm{yr^{-1}}$. A central innovation is the use of the global biogeochemical Argo database to assign carbon subduction probabilities regionally, which reveals where the pump is most active. Primary hotspots (probability of carbon subduction more than 10%) emerge in the subpolar North Atlantic, Drake Passage, and along the Pacific–Antarctic Ridge, while secondary hotspots (probability of carbon subduction between 2.5% and 10%) appear in the Agulhas Current, Kerguelen Plateau, and tropical Indian Ocean. Most export is sustained by springtime activity in regions of high eddy kinetic energy, especially in the Southern Ocean and subpolar North Atlantic. The locations of carbon subduction hotspots broadly align with earlier regional studies in the Southern Ocean (M. L. Chen & Schofield, 2024; Lacour et al., 2023; Llort et al., 2018), North Atlantic (A. R. Johnson & Omand, 2021), Kuroshio (S. Chen et al., 2021), Mediterranean (Mahadevan et al., 2020), and California Current (Stukel et al., 2017, 2018), while also highlighting previously unreported hotspots in the tropical Indian and Pacific Oceans. The magnitude of the eddy subduction pump found in this study (0.05 [<0.01–0.28] Pg C $\mathrm{yr^{-1}}$) is remarkably consistent with a recent global, residual-based estimate of the eddy subduction pump derived from a 4-dimensional particulate organic carbon product, which yields an export of ~0.06 Pg C $\mathrm{yr^{-1}}$ (Bellacicco et al., 2025), despite relying on an entirely different methodological framework.

4.2. Eddy Subduction Pump Peaks in Spring

The eddy subduction pump peaks in spring in well-sampled extratropical regions, including the North Atlantic and the Southern Ocean, whereas no significant seasonality is detected in the tropics. The absence of seasonality in the tropics likely reflects weaker submesoscale activity (Dong et al., 2021), lesser seasonality of the mixed layer depth and the water column stratification (Kara et al., 2003; Sérazin et al., 2023) and a relative lack of phytoplankton blooms (Agarwal et al., 2024; Callies et al., 2016). In contrast, the spring peak in the North Atlantic and the Southern Ocean is consistent with Erickson and Thompson (2018), who found that eddy subduction in the Northeast Atlantic occurs over a narrow late winter to early spring window, when strong physical instabilities coincide with elevated surface carbon content. This result, however, challenges other work suggesting that carbon subduction observed in 7 Argo floats in the subpolar North Atlantic peaks in summer (A. R. Johnson & Omand, 2021). Surprisingly, the North Pacific shows no statistically significant seasonality, despite a noticeable, yet undersampled, spike in the Fall. Sparse Argo data likely mask any seasonality that might exist in the Kuroshio region or elsewhere in the subpolar Pacific.

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4.3. Subduction Hotspots Found in High EKE Regions

There is an ongoing discussion regarding the generating processes at play behind physical subduction events. Physical eddy subduction events are by-products of the restratification of submesoscale fronts (Lévy et al., 2024; Taylor & Thompson, 2023), but the mechanisms governing front formation and restratification of these fronts remain poorly understood and debated. As we mentioned in the introduction and results sections, fronts may emerge either because of spontaneous frontogenesis or forced frontogenesis (Blumen, 2000; Shakespeare & Taylor, 2013). Spontaneous frontogenesis is driven by inertial adjustment of an initially unbalanced flow and can develop in a matter of hours (Blumen, 2000; Shakespeare & Taylor, 2013). In contrast, forced frontogenesis results from persistent mesoscale strain in regions of high balanced EKE and typically unfolds over days to weeks (Hoskins & Bretherton, 1972; Shakespeare & Taylor, 2013).

The positive association between hotspots of physical subduction and of balanced EKE in the extratropics is consistent with a slow, forced frontogenesis pathway: persistent mesoscale strain builds intense, long-lived fronts whose secondary circulations are strong enough to penetrate well beneath the export horizon. In contrast, balanced EKE is negatively correlated with subduction at tropical latitudes, suggesting that a faster, spontaneous adjustment energized by inertial oscillations, might dominate there. If in the tropics spontaneous frontogenesis dominates, its shorter time scale, on the order of hours (Shakespeare & Taylor, 2013), makes it extremely difficult to simulate in models and could partially explain why even high-resolution biogeochemical models show much weaker activity of the eddy subduction pump in the tropics than in the extratropics (Resplandy et al., 2019).

The consistent and robust negative association between unbalanced EKE and the probability of physical subduction in both the tropics and the extratropics might suggest that vigorous internal tides and internal gravity waves siphon energy away from the balanced mesoscale strain field, flattening fronts and suppressing both forced and spontaneous frontogenesis (Shakespeare & Taylor, 2013; Thomas, 2017). This competition of internal tides and gravity waves with physical subduction could explain why fewer sustained subduction events are observed when unbalanced EKE is high.

4.4. Contributions From Symmetric and Mixed Layer Instabilities

The restratification of submesoscale fronts is due to physical instabilities such as symmetric instabilities (Bachman et al., 2017; Thomas et al., 2013) or mixed layer instabilities (Boccaletti et al., 2007; Fox-Kemper et al., 2008) or, most often, a conjunction of both (Stamper & Taylor, 2017).

We present statistical associations indicating that detected subduction events preferentially occur under weakly stratified upper-ocean conditions, which are consistent with environments favorable to symmetric instability. These statistical associations are consistent with earlier observational and modeling studies suggesting that symmetric instability can contribute substantially to submesoscale restratification alongside mixed layer instabilities (M. L. Chen & Schofield, 2024; Erickson & Thompson, 2018). Several regions identified in this study coincide with areas where symmetric instability has been diagnosed in high-resolution simulations, suggesting a potential role for symmetric instabilities in shaping the observed subduction patterns (Dong et al., 2021).

These considerations are important because the type of instability driving physical and carbon subduction has important implications for how the eddy subduction pump can be represented in biogeochemical models. Whereas mixed layer instabilities are typically resolved at a horizontal resolution of about 1 km (Boccaletti et al., 2007), symmetric instabilities involve meter-scale processes (Dong et al., 2021). As a result, if symmetric instabilities play a substantial role, their explicit representation remains challenging for present-day global climate models and would likely require either major increases in resolution or improved parameterizations (Dong et al., 2021; Stamper & Taylor, 2017).

Our argument in favor of symmetric instabilities rather than mixed layer instabilities deserves scrutiny. Indeed, this argument is statistical rather than mechanistic, and both N^2 and MLD are estimated at the time of the subduction event's detection by the Argo float, rather than at the time the event occurred. The time lag between the subduction and the detection might be up to 3 months, especially in particularly quiet, low mean-flow and low EKE regions (e.g., Southern Rockall Trough) as it was shown by A. R. Johnson and Omand (2021). Therefore, the mixed layer depth and water column stratification at time of detection might be quite different from what they were at time of subduction. Additionally, the fact that we only look at subduction events beyond 200 m might

imply that we miss subduction events, as it was shown by Mahadevan et al. (2020) that subduction driven by mixed layer instabilities in the Mediterranean was very shallow, occurring in the top 150 m of the ocean. We therefore interpret this statistic argument on symmetric versus mixed layer instabilities as suggestive rather than definitive, and stress that targeted process studies and high-resolution regional simulations are essential to better understand the relative roles of these instabilities in generating subduction events.

4.5. Limitations

The main limitations hindering the investigation of the eddy subduction pump through Argo floats are the insufficient spatial and temporal coverage of the Argo database. The chronic undersampling of the Pacific Ocean, the Tropics, and the Arctic Ocean (see Section 2.1) makes it hard to give any estimates of the magnitude and seasonality of the eddy pump in those regions. For instance, the absence of a seasonal carbon subduction hotspot in the Kuroshio came as a surprise. However, it is hard to determine whether this absence is due to a generally lower biological productivity of the North-Western Pacific Ocean relative to the North Atlantic, as was shown by Lee et al. (1981), Kimura (1997), and Kasai et al. (2002) or insufficient coverage of the region by Argo floats (see Figure S1 in Supporting Information S1). On the other hand, the particularly well-sampled hotspot downstream of the Kerguelen may have led Llort et al. (2018) to overestimate the importance of this hotspot of carbon subduction relative to other hotspots in the Southern Ocean, near the East-Pacific Antarctic Ridge or downstream of the Drake Passage.

An additional limitation is the lack of information on vertical velocities of carbon subduction and the age of carbon subduction which are required to infer the eddy subduction POC export flux. In this study we used a range of characteristic subduction velocities, $W \in [20, 500]$ m d^{-1}, and apply it uniformly to all detected events. Because the local daily export flux (Q) scales linearly with the vertical velocity (W), see Equation 3, the one order of magnitude spread in the vertical velocity propagates directly into an order of magnitude uncertainty in our global export estimate. Ideally, one would estimate a velocity specific to each individual filament, but such information cannot be recovered from Argo profiles alone. Furthermore, by fixing a global export horizon of 200 m, we may have missed superficial carbon subduction events confined to the mixed layer, as is the case for the Mediterranean Sea (Mahadevan et al., 2020).

Event age adds further uncertainty. The POC anomaly observed by a float is only the residue of the original export, remineralization and lateral diffusion erode the signal between subduction and detection by the Argo float. We correct for this loss by assuming a maximum lag of 3 months and an e-folding decay time of 90 days, following A. R. Johnson and Omand (2021). While appropriate for relatively quiescent basins, this correction is likely conservative in energetic regions such as the Antarctic Circumpolar Current, where coherent POC anomalies dissipate more rapidly. In such settings a larger multiplicative factor would be required to recover the initial export. Taken together, the uncertainties in both the timing of observation and the choice of W expand the plausible range of the eddy-subduction export by roughly two orders of magnitude.

In addition, Argo floats are unable to measure dissolved organic carbon (DOC), effectively only measuring a fraction of the downward flux of carbon. Eddy-driven subduction has been recognized to export DOC along with POC (Omand et al., 2015; Resplandy et al., 2019), and may even be exporting nearly as much DOC as POC (55% of POC and 45% of DOC) (Poupon et al., 2025; Resplandy et al., 2019). Argo floats are unable to measure DOC and therefore provide an incomplete view of carbon export by the eddy subduction pump. Moreover, Argo floats do not detect upward fluxes of organic carbon that could partially offset the contribution of carbon subduction events to the eddy subduction pump (Resplandy et al., 2019). In fact, all exports reported here are gross downward fluxes. We do not quantify compensating upwards fluxes, and therefore we do not interpret the export fluxes estimates as net carbon sequestration or as an offset of anthropogenic emissions.

4.6. Implications for the Global Biological Carbon Pump

Our global estimate of the annual export flux of POC by the eddy subduction pump is 0.05 [<0.01–0.28] Pg C yr^{-1}. This estimate sits at the low end of the [0–2] Pg C yr^{-1} global estimate of (Boyd et al., 2019), based on extrapolations of Llort et al. (2018)'s Southern Ocean study to the globe, but closely matches a recent global estimate of ~0.06 Pg C yr^{-1} obtained from a 4-dimensional POC product (Bellacicco et al., 2025). While Bellacicco et al. (2025) infer the contribution of the eddy pump to total carbon export as a residual between large-

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scale and total export rather than through explicit event detection, the close agreement in magnitude provides support for our estimate. Our estimate of the magnitude of the eddy subduction pump represents less than 5% of the total export of POC by the biological pump (~10 Pg C yr^{-1}; Boyd et al. (2019); Nowicki et al. (2022); Wang et al. (2023)). Incorporating dissolved organic carbon using the ~55% POC/45% DOC partitioning from (Poupon et al., 2025) would raise our estimate to 0.10 [<0.01–0.51] Pg C yr^{-1}. Even then, the eddy subduction pump still represents less than 5% of the total export of organic carbon, which is ~12 Pg C yr^{-1} including POC and DOC (Boyd et al., 2019), in agreement with (Resplandy et al., 2019). Moreover, we estimate that only 25% of exported POC enters waters with reemergence times larger than 50 years, consistent with rapid remineralization of physically exported POC found by Poupon et al. (2025). Overall, the relatively small contribution of the eddy subduction pump suggests that it can likely be neglected for first-order estimates of the total biological carbon export. This is good news, given that current climate models lack the resolution to capture the effects of the eddy subduction pump on the global carbon budget.

5. Conclusion

We present a framework for the global detection of the eddy subduction pump, finding springtime hotspots in the Southern Ocean and the North Atlantic associated with high eddy kinetic energy. We estimated the POC export flux attributable to the eddy subduction pump to 0.05 Pg C yr^{-1} with a likely range (<0.01–0.28 Pg C yr^{-1}). This estimate may be updated when more Argo float data becomes available. We present some evidence that the eddy subduction pump is due to an interplay of mixed layer and symmetric instabilities, which are processes that happen at scales which are beyond the reach of global biogeochemical ocean models and Earth system models. However, the small magnitude of the export rate of the subduction pump, less than 5% of the total export of particulate organic carbon by the biological pump, seem to suggest that the adequate representation of the eddy subduction pump in ocean models is not a priority for reducing our uncertainty on the global ocean carbon budget.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Float data were downloaded from the USGODAE archive; https://argo.ucsd.edu/data/data-from-gdacs/ using the OneArgo-R toolbox https://github.com/NOAA-PMEL/OneArgo-R (Cornec et al., 2022), developed by NOAA-PMEL and licensed under GNU General Public License v3.0 (GPL-3, 29 June 2007). Our analysis used delayed-mode, quality-controlled data retrieved on 6 May 2024.

All analyses were conducted in R (version 4.4.1; released 14 June 2024), which is distributed under GNU GPL v3, with the tidyverse suite used for data manipulation and visualization (R Core Team, 2024; Wickham et al., 2019). Statistical modeling and generalized additive models (GAMs) fittings employed the mgcv package (S. Wood, 2000; S. N. Wood, 2017). Thermodynamic variables were computed using TEOS-10 implementations in the gsw package (Kelley et al., 2024; McDougall & Barker, 2011). The code associated with this manuscript for data processing and analysis, including an implementation of the detection algorithm in R is licensed under GNU GPL v3 and published on Zenodo (Keutgen De Greef, 2025).

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